

# QAZI MAHAD AHMAD

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Peshawar, Pakistan

## PROFESSIONAL SUMMARY

Machine Learning Engineer with hands-on experience designing, developing, and deploying end-to-end machine learning solutions. Proficient in Python, Scikit-learn, SQL, and data analysis, with expertise in data preprocessing, exploratory data analysis, feature engineering, model development, evaluation, and model deployment. Experienced in building predictive models and leveraging data-driven insights to solve real-world business problems. Passionate about developing scalable, high-performance machine learning systems while continuously expanding expertise in modern AI and machine learning technologies.

## TECHNICAL SKILLS

**Programming Languages:** Python, C+

**Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, XGBoost, Feature Engineering, Model Evaluation

**Data Science & Analytics:** NumPy, Pandas, Exploratory Data Analysis (EDA), Data Cleaning, Data Preprocessing

**Data Visualization:** Matplotlib, Seaborn, Plotly

**Web & APIs:** Streamlit, FastAPI, REST APIs, BeautifulSoup4, Requests

**Databases:** SQL, MySQL, DuckDB

**Version Control & Development:** Git, GitHub, VS Code, Jupyter Notebook, PyCharm

**AI & ML Platforms:** HuggingFace

**Operating Systems & Tools:** Ubuntu Linux, Windows, Bash, Docker

## WORK EXPERIENCE

### Machine Learning Engineering Intern | FlyRank AI

July 2026 – Present

- Contribute to the end-to-end machine learning pipeline, including data preprocessing, feature engineering, model development, evaluation, and experimentation to build data-driven solutions.

### Machine Learning Engineer | Independent

- Designing end-to-end ML pipelines, including data preprocessing, feature engineering, model training, evaluation, and deployment using Scikit-learn, TensorFlow, and PyTorch.

## PROJECTS

### Digit Recognition System (Digit.AI) | Machine Learning & Deployment Project

**Live:** [huggingface.co/spaces/Mahad0007/digit-recognition](https://huggingface.co/spaces/Mahad0007/digit-recognition) | **GitHub:** [github.com/QaziMahadAhmad/digit-recognition](https://github.com/QaziMahadAhmad/digit-recognition)

- Built a complete end-to-end ML pipeline that recognises handwritten digits (0–9) in real time with 95.03% test accuracy on 70,000 MNIST samples.
- Applied StandardScaler and PCA to compress 784 pixel features to 331 dimensions (keeping 95% of variance), solving the curse of dimensionality for KNN.
- Used 5-fold stratified cross-validation across K values (1–25) to select the optimal KNN classifier (K=3, Euclidean distance).
- Built a FastAPI backend (single Python file no templates) with an embedded HTML/CSS/JS UI supporting canvas drawing and image upload.
- Deployed live on Hugging Face Spaces using Docker; models stored in a separate HF model repo and downloaded at build time.

### Email Spam Detection System | Machine Learning Project

**Live:** [email-spam-detector-09.streamlit.app](https://email-spam-detector-09.streamlit.app) | **GitHub:** [github.com/QaziMahadAhmad/digit-recognition](https://github.com/QaziMahadAhmad/digit-recognition)

- Developed a text classification model to identify spam and non-spam emails using the Naive Bayes algorithm.
- Applied NLP techniques including tokenization, stopword removal, and TF-IDF vectorization for feature extraction.
- Improved model performance using Scikit-learn, achieving strong accuracy on real-world email data.

## Customer Churn Prediction System | Machine Learning & FastAPI Project

**GitHub:** <https://github.com/QaziMahadAhmad/churn-predictor>

- Built an end-to-end machine learning pipeline for telecom customer churn prediction, including data preprocessing, feature engineering, model training, and evaluation.
- Compared six classification models and selected the best-performing model through hyperparameter tuning and cross-validation.
- Developed a FastAPI REST API with an interactive HTML/CSS/JavaScript dashboard for real-time churn prediction.
- Applied SHAP and feature importance for model explainability and packaged the application with Docker for deployment.

## Math Score Predictor | Machine Learning & Web App Project

**GitHub:** [https://github.com/QaziMahadAhmad/math\\_score\\_prediction](https://github.com/QaziMahadAhmad/math_score_prediction)

- Built a Linear Regression model to predict student Math scores based on Reading and Writing scores.
- Developed and deployed an interactive web application using Streamlit for real-time predictions.
- Evaluated model performance using  $R^2$ , MAE, and RMSE, and visualized results with actual vs predicted plots.

## CERTIFICATIONS

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- Machine Learning Specialization | DeepLearning.AI & Stanford University | Coursera  
Link : <https://coursera.org/share/a79750d18e68d39b57d0a51d37f8887a>
- Mathematics for Machine Learning and Data Science Specialization | DeepLearning.AI | Coursera  
Link : <https://coursera.org/share/a79750d18e68d39b57d0a51d37f8887a>
- AI fluency : Framework & Foundation | Antropic  
Link : [https://drive.google.com/file/d/1kw\\_KOWfSD0fxv5093qGbxg03m36imXEg/view?usp=sharing](https://drive.google.com/file/d/1kw_KOWfSD0fxv5093qGbxg03m36imXEg/view?usp=sharing)
- Programming for Everybody | University of Michigan | Coursera  
Link : <https://coursera.org/share/a79750d18e68d39b57d0a51d37f8887a>

## EDUCATION

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### Bachelor of Science in Computer Science (BSCS)

2024 – 2028

University of Peshawar

### Higher Secondary School Certificate (HSSC), Pre-Engineering

2022 – 2024

Government College Peshawar

Grade : A

## ADDITIONAL

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**Areas of Interest:** Generative AI, LLMs, RAG, AI Agents, NLP, Computer Vision

**Professional Strengths:** Analytical Problem-Solving, Critical Thinking, Self-Directed Learning, Curiosity, Experimentation, Adaptability, Collaboration

**Languages:** English (Professional), Urdu (Native), Pashto (Native)